Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	View XAI Explanations
2. Abbreviated Title	View XAI
3. Use Case Id	UC-04
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	After a prediction, the Clinician views Explainable AI visualizations that explain why the model made a specific prediction, using three independent XAI methods (Grad-CAM, SHAP, Attention Weights) and their inter-method agreement metrics.
5.1. Pre Conditions	1. A prediction has been successfully completed (UC-03). 2. XAI computation modules (Grad-CAM, SHAP, Attention) are available.
5.2. Task Sequence	1. System invokes UC-12: Compute Grad-CAM to generate the heatmap. 2. System invokes UC-13: Compute SHAP Values for feature importance. 3. System invokes UC-14: Extract Attention Weights for attention distribution. 4. System invokes UC-18: Compute XAI Agreement Metrics. 5. System renders all visualizations in Plotly.js chart containers. 6. Clinician interacts with charts (hover, zoom, filter features).
5.3. Post Conditions	1. Grad-CAM heatmap, SHAP bar chart, Attention visualization are rendered. 2. XAI agreement score is displayed. 3. Clinician understands why the model made the prediction.
6. Modification History	Date: September 11, 2026
7. Author	Dhruv Singla (1024170434), Srishti (1024170451)